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When the Link Is the Variable: Delay-Tolerant Networking and Bundle Custody Transfer at the C5ISR Edge

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Abstract

Conventional networking treats the link as a constant: present, available on demand, reliable within its designed failure envelope, and amenable to retransmission when packets are lost. This assumption pervades transport protocols, application architecture, and operational doctrine from the enterprise datacenter to the forward-deployed tactical node. It is an assumption that the tactical edge systematically violates. At the C5ISR edge under DDIL conditions — Disrupted, Disconnected, Intermittent, Limited-bandwidth — the link is not a constant. It is a variable with a distribution of availability that must be characterized, managed, and exploited architecturally rather than assumed away operationally.

This paper develops the architectural argument that treating the link as a variable demands a transport architecture fundamentally different from the connection-oriented model. The transport primitive is not a connection that carries data; it is a custody transfer that moves state across link windows. When the link is present, custody moves. When the link is absent, state is held — durably, verifiably, and with governance integrity — until the link returns. The Bundle Protocol and Delay-Tolerant Networking (DTN) architecture, originally developed for interplanetary communication, provide the engineering specification for this custody-based transport model. Applied to the tactical edge, they constitute the first substrate primitive that all subsequent operational guarantees depend upon.

The paper argues, further, that transport custody is not merely a networking mechanism — it is a governance primitive. The decision of what gets custody priority, which nodes act as custodians, and what constitutes a custody failure requiring human escalation are governance decisions that the HGC³AE² framework must address before the transport architecture can be operationally sound. A custody-based transport layer that operates without governance is a store-and-forward system. A custody-based transport layer governed by the H half, curated by the C³ half, and enforced by the AE² half is the transport substrate of a tactically coherent C5ISR architecture.

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ABSTRACT

When the link is a constant — high-bandwidth, low-latency, always-on — the agentic operating model can treat communication as transparent infrastructure. When the link is **the variable** — intermittent, asymmetric, partition-prone — communication becomes a first-class architectural concern. This paper argues that delay-tolerant networking (DTN) and bundle-custody protocols provide the operational substrate for agentic operation under variable-link conditions, and develops what custody-transfer semantics mean when the bundles carry governance state, agentic work-in-flight, and persistence-tier writes.¹² The argument is that agentic systems on the tactical substrate (EDO-P1) require a communication architecture that does not assume the link, and that DTN's bundle protocol — specifically Bundle Protocol v7 (BP v7) with Bundle Protocol Security (BPsec) — is the standardized operational substrate for this requirement.³⁴

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1. INTRODUCTION

The cloud-native communication model presumes the link: TCP/IP over reliable infrastructure, with retransmission absorbing transient loss and end-to-end semantics preserving application invariants.⁵ On the tactical substrate (EDO-P1), this presumption fails. Links are intermittent — sometimes for seconds, sometimes for hours, occasionally for days. Bandwidth is asymmetric and constrained. Latency varies by orders of magnitude. End-to-end paths may not exist for substantial fractions of the operating window.⁶⁷

DTN was developed to address exactly this. The architectural commitment of DTN — store-and-forward at the application layer, with custody transfer between intermediate nodes — produces a communication model in which the end-to-end path is virtualized over a sequence of opportunistic hops, each with custody discipline that ensures eventual delivery without requiring

1 Justin H. Kuiper, CISSP, *HGC³AE² Framework* (v1.0-preprint, 2026), zenodo.org/records/19869285.

2 Vinton G. Cerf et al., “Delay-Tolerant Networking Architecture,” RFC 4838, IETF, 2007, <https://datatracker.ietf.org/doc/rfc4838/>.

3 Scott C. Burleigh, Kevin Fall, and Edward J. Birrane III, “Bundle Protocol Version 7,” RFC 9171, IETF, 2022, <https://datatracker.ietf.org/doc/rfc9171/>.

4 Edward J. Birrane III and Kenneth McKeever, “Bundle Protocol Security (BPsec),” RFC 9172, IETF, 2022, <https://datatracker.ietf.org/doc/rfc9172/>.

5 Martin Kleppmann, *Designing Data-Intensive Applications* (Sebastopol, CA: O'Reilly, 2017).

6 Vinton G. Cerf et al., “Delay-Tolerant Networking Architecture,” RFC 4838, IETF, 2007, <https://datatracker.ietf.org/doc/rfc4838/>.

7 Kevin Fall, “A Delay-Tolerant Network Architecture for Challenged Internets,” *Proceedings of ACM SIGCOMM '03*, 2003, <https://doi.org/10.1145/863955.863960>.



continuous connectivity.⁸ For agentic operation, the bundle protocol carries the cargo that the operating model needs to move across the variable link: governance envelope updates, agentic work-in-flight, persistence-tier writes, reconciliation merges, and observability telemetry.

The paper proceeds in canonical Mission-Ready shape. §2 names five failure modes of cloud-native communication on variable links. §3 specifies bundle custody for agentic cargo. §4 treats custody as continuous. §5 binds DTN to Skipjack mechanisms. §6 returns to HGC³AE² and frames the role of variable-link communication in mission-ready operation.

2. FAILURE MODES OF CLOUD-NATIVE COMMUNICATION ON VARIABLE LINKS

End-to-end timeout. Cloud-native protocols (HTTPS, gRPC, mTLS) presume end-to-end paths; when paths are intermittent, sessions time out and applications halt.⁹

Retry-storm amplification. Application-level retries amplify transient outages into retry storms that consume scarce bandwidth and accelerate substrate degradation.

State loss on partition. Work-in-flight that lacks custody transfer is lost when partitions occur; the sender treats it as delivered, the receiver never sees it.

Stale governance propagation. Governance envelope updates that cannot reach the substrate during partition produce drift between authoritative governance and operating envelope.

Observability gap. Telemetry that cannot transmit during partition either buffers locally (consuming scarce capacity) or is dropped (creating reconstruction gaps in the operator's view).

3. BUNDLE CUSTODY FOR AGENTIC CARGO

The bundle protocol's central commitment is **custody transfer**: at each intermediate node, the node accepts custody of the bundle, persists it locally, and acknowledges custody to the previous node. The previous node may then release the bundle from its own custody. End-to-end delivery is virtualized over the sequence of custody transfers.¹⁰¹¹

For agentic cargo, three bundle classes are load-bearing:

Governance bundles. Envelope updates, policy changes, persona registry updates. Governance bundles require integrity attestation (BPsec) and explicit authority validation at each custody point. Tampering with governance bundles is a security event of the highest severity.¹²

⁸ Kevin Fall and Stephen Farrell, "DTN: An Architectural Retrospective," *IEEE Journal on Selected Areas in Communications* 26, no. 5 (2008): 828–836, <https://doi.org/10.1109/JSAC.2008.080609>.

⁹ Vinton G. Cerf et al., "Delay-Tolerant Networking Architecture," RFC 4838, IETF, 2007, <https://datatracker.ietf.org/doc/rfc4838/>.

¹⁰ Scott C. Burleigh, Kevin Fall, and Edward J. Birrane III, "Bundle Protocol Version 7," RFC 9171, IETF, 2022, <https://datatracker.ietf.org/doc/rfc9171/>.

¹¹ Kevin Fall, "A Delay-Tolerant Network Architecture for Challenged Internets," *Proceedings of ACM SIGCOMM '03*, 2003, <https://doi.org/10.1145/863955.863960>.

¹² Edward J. Birrane III and Kenneth McKeever, "Bundle Protocol Security (BPsec)," RFC 9172, IETF, 2022, <https://datatracker.ietf.org/doc/rfc9172/>.



Work bundles. Agentic work-in-flight: tasks, intermediate state, decision records. Work bundles require deduplication discipline (a bundle delivered multiple times by alternative paths must be applied once) and explicit completion semantics.

Persistence bundles. Writes to the authoritative tier originating from disconnected substrates. Persistence bundles require ordering discipline within the reconciliation window and explicit conflict-resolution semantics at the authoritative tier (typically CRDT-mediated, per EDO-P4).¹³

Each bundle class carries BPsec attestation. The substrate verifies attestation at custody acceptance; bundles failing attestation are not propagated.

4. CUSTODY AS A CONTINUOUS OPERATIONAL PROCESS

Custody operates continuously through three phases. **Acceptance** verifies the bundle's attestation and accepts custody. **Propagation** forwards the bundle to the next hop when a path is available. **Release** discharges custody when downstream acknowledgment is received. The cycle runs at every intermediate node continuously; bundles in transit at any moment have multiple custody-points participating in their delivery.

For the tactical substrate, the cycle accommodates partition explicitly: bundles persisted in custody at a substrate that loses connectivity remain in custody until either downstream connectivity is restored (normal propagation) or a custody timeout returns custody to the previous node (recovery).

5. SKIPJACK COUPLING: DTN CARGO AS ENFORCEMENT SURFACE

Skipjack mechanisms operate over DTN cargo with adaptations.¹⁴ **Context integrity** is enforced through BPsec attestation at every custody point. **Structured routing** operates over the bundle space, not over a continuous communication substrate. **HITL control points** fire at custody-timeout events that signal partition prolonged enough to warrant human intervention. **Iterative feedback** closes asynchronously through bundle-mediated telemetry returning from disconnected substrates.

The coupling preserves Skipjack's commitments under variable-link conditions; the architectural cost is the explicit treatment of bundle cargo and custody semantics rather than the implicit reliance on connection-oriented end-to-end communication.

6. IMPLICATIONS AND CONCLUSION

The implications of treating the link as the variable extend to architectural investment (BP v7 and BPsec are operational substrates, not optional add-ons), operational discipline (custody management is a runtime concern), and adoption (organizations deploying to the tactical substrate must architect for DTN explicitly).

The core claim: **DTN bundle custody is the standardized operational substrate for agentic communication when the link is variable; without it, the cloud-native communication model produces structural failure modes on the tactical substrate.**

Returned to HGC³AE² letter by letter:

¹³ EDO-P4 *State Coherence Problem* promotion (in concurrent drafting; yks-build#96).

¹⁴ Justin H. Kuiper, CISSP, *Skipjack Protocol* (v1.0-preprint, 2026), zenodo.org/records/19869293.



H — Human-governed. Governance bundles carry envelope updates with attestation; the substrate cannot operate against unattested governance.

G — Curated Context. Curated context propagates via persistence bundles with explicit ordering and reconciliation discipline.

C¹ — Cybersecurity. BPsec at every custody point; tamper-evident bundle protocol.

C² — Continuity. Custody transfer is the architectural form of continuity over variable links.

C³ — Coherence. Reconciliation merges at the authoritative tier; CRDT discipline preserves merge coherence (EDO-P4).

A — Agentically Engineered. Bundle propagation is agentic within the governance envelope encoded in the bundles themselves.

E — Executed. Custody cycle executes continuously at every node.

The exponent — the closure — is the bundle-mediated telemetry loop that informs envelope updates that propagate back as governance bundles.

The remaining EDO papers develop the variable-link surface in adjacent dimensions: state coherence (EDO-P4), out-of-path supervision (EDO-P5), protocol selection (EDO-P6).¹⁵¹⁶¹⁷

REFERENCES

— end v1.0-preprint-draft —

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15 EDO-P4 *State Coherence Problem* promotion (in concurrent drafting; yks-build#96).

16 EDO-P5 *Orchestration Problem* promotion (in concurrent drafting; yks-build#97).

17 EDO-P6 *Protocol Selection Problem* promotion (in concurrent drafting; yks-build#105).